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EXAMINER	
DIXON, ANNETTE FREDRICKA	

ART UNIT	PAPER NUMBER
3785	

NOTIFICATION DATE	DELIVERY MODE
01/18/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/959,368	Applicant(s) BARLOW et al.	
	Examiner ANNETTE F DIXON	Art Unit 3785	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/4/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-21 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-21 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/4/2022 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☑ Certified copies of the priority documents have been received in Application No. 17426362.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 10/4/2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Primary Examiner acknowledges Claims 1-21 are pending in this application, as originally filed on October 4, 2022.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Drawings

3. The drawings are objected to as failing to comply with 37 CFR 1.84(p)(4) because:

a. Reference character "3000" has been used to designate "patient interface", "nasal cradle interface", and "non-invasive patient interface". Appropriate correction is required.

b. Reference character "3100" has been used to designate "forming structure", "seal-forming structure", and "sealing structure". Appropriate correction is required.

c. Reference character "3106" has been used to designate "naso-labial sulcus area", and "naso-labial sulcus engagement area". Appropriate correction is required.

d. Reference character "3107" has been used to designate "flanges", and "naso-labial sulcus engagement area". Appropriate correction is required.

e. Reference character "3118" has been used to designate "membrane layer", "material membrane layer", "cured membrane layer", "thicker membrane layer", and "thinner membrane layer". Appropriate correction is required.

f. Reference character "3354" has been used to designate "outer layer", "lower layer", "opposite outer layer" and "rigid outer layer". Appropriate correction is required.

g. Reference character "3363" has been used to designate "inferior tube portion" and "tube". Appropriate correction is required.

h. Reference character "9322" has been used to designate "eyelets" and "slit". Appropriate correction is required.

i. Reference character "9360" has been used to designate "crown connector" and "headgear tubes". Appropriate correction is required.

4. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

5. The disclosure is objected to because of the following informalities:

j. Reference character "3000" has been used to designate "patient interface", "nasal cradle interface", and "non-invasive patient interface". Appropriate correction is required.

k. Reference character "3100" has been used to designate "forming structure", "seal-forming structure", and "sealing structure". Appropriate correction is required.

l. Reference character "3106" has been used to designate "naso-labial sulcus area", and "naso-labial sulcus engagement area". Appropriate correction is required.

m. Reference character "3107" has been used to designate "flanges", and "naso-labial sulcus engagement area". Appropriate correction is required.

n. Reference character "3118" has been used to designate "membrane layer", "material membrane layer", "cured membrane layer", "thicker membrane layer", and "thinner membrane layer". Appropriate correction is required.

o. Reference character "3354" has been used to designate "outer layer", "lower layer", "opposite outer layer" and "rigid outer layer". Appropriate correction is required.

p. Reference character "3363" has been used to designate "inferior tube portion" and "tube". Appropriate correction is required.

q. Reference character "9322" has been used to designate "eyelets" and "slit". Appropriate correction is required.

r. Reference character "9360" has been used to designate "crown connector" and "headgear tubes". Appropriate correction is required.

Claim Rejections - 35 USC § 112

6. The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

7. Claims 1-21 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

As to Claim 1, Applicant recites the features of the gas delivery tube having a "first elongate length of material", a "second elongate length of material", and a "third elongate length of material", whereby the "first elongate length of material" is joined to the "second elongate length of material" in order to form a "first seam", and the "second elongate length of material" is joined to the "third elongate length of material" in order to form a "second seam"; however, the structure and relationship of the manner by which the "third elongate length of material" is joined to "first elongate length of material" is missing from the claim listing.

Turning to the instant claim listing, the claim limitations state the resultant effect of the first and second seam is the construction "to form the tube wall of the gas delivery tube" (Claim

1, Second to Last Stanza); thus, the questions remains how can a tube wall be formed when the structure is not enclosure in such a manner to operate for its intended purpose to provide gas delivery. The failure to provide the closure of the "third elongate length of material" joined to "first elongate length of material" effectively provides an annular tube wall with a sealing flap. Thus, **Primary Examiner is unclear how and the manner by which the gas delivery tube is constructed.**

More explicitly, to show the range of prior art available by this indefinite claim language:

Does the "third elongate length of material" join to "first elongate length of material" in order to provide an annular cross section of the tube wall of the gas delivery tube - similar to Grashow (11,273,279)?

OR ALTERNATIVELY, Does the "third elongate length of material" remain separate and distinct structurally from the "first elongate length of material" to provide an in-series configuration of the gas delivery tube whilst still fluidly connected - similar to Rinaldi (2006/0180151)?

STILL FURTHER, Does the "third elongate length of material" provide a sealing flap or intended to roll over the tube wall as created by the connection of the first and second seams to protect the first and second seams - similar to Ferrand (2005/0069666)?

Turning to the original disclosure as filed, it appears the feature Applicant is attempting to seek for patent protection is shown in Figures 32 and 33, whereby the combination of each of the "first elongate length of material", the "second elongate length of material", and the

"third elongate length of material" form the cross section of the tube wall of the gas delivery tube having three seams. This structure appears to be confirmed by the recitation in original Claim 13, which recites the orientation of the third elongate length of material" is joined to "first elongate length of material" forming the "third seam".

Applicant is reminded, *although the claims are interpreted in light of the specification, limitations from the specification are not read into the claims*. See *In re Van Geuns*, 988 F.2d 1181, 26 USPQ2d 1057 (Fed. Cir. 1993). In this case, the examination of the claims as pending does not effectively particularly point out and distinctly claim the subject matter which the inventor(s) deem as their invention. This chasm between the claimed invention and the disclosed invention gives way for the introduction of prior art such as the aforementioned Rinaldi (2006/0180151) in series configuration, Ferrand (2005/0069666) in varying dimensions of the tube wall thickness along the elongated length of material, and still further gives way for additional, unrecited elements by the use of the transitional phrase "comprising" (MPEP § 2111.03) such as the structure of aforementioned Grashow (11,273,279) which includes additional elongate lengths of material beyond the explicitly claimed "first elongate length of material", "second elongate length of material", and "third elongate length of material" in order to form the cross section of the tube wall of the gas delivery tube having three seams.

In the interest of advancing prosecution of this application, at the bare minimum Applicant should: 1) incorporate the limitations of Claim 13 into independent Claim 1, to effectively describe the structure and relationship of each of the elongate length of material to form the gas delivery tube; 2) further, it would behoove Applicant to clearly and unmistakable recite the configuration as shown in Figures 32 and 33 wherein each of the elongate lengths of

material form the cross section of the tube wall of the gas delivery tube; 3) finally, it would behoove Applicant to modify the transitional phrase of "consisting" (MPEP § 2111.03) - which "excludes any element, step, or ingredient not specified in the claim."

The actions of each of these three (3) actions would be helpful to effectively provide a chasm between the prior art made of record and the claimed invention - *although further search and consideration will still be required prior to an indication of allowable subject matter*. Dependent Claims 2-12, and 19-21 incorporate the indefinite subject matter from which they depend. Appropriate correction and clarification is required.

As to Claim 18, the limitation "the rigidizing element"; lacks insufficient antecedent basis for this limitation in the claim. From a review of the claim listing, the feature of the "rigidizing element" was initially introduced in Claim 17; thus, it appears the dependency of Claim 18 should be to Claim 17 and NOT to Claim 11. Appropriate correction and clarification is required.

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. Claims 1, 2, 9-14, and 19-21 are rejected under 35 U.S.C. 103 as being unpatentable over Grashow (11,273,279 - 371 national stage entry of PCT/IB2017/056005 having a PCT publication date of April 12, 2018) in view of Ovizinsky et al. (2014/0102456) - *whereby each elongate length of material provides an annular cross section of the tube wall of the gas delivery tube.*

As to Claims 1 and 19, Grashow discloses a positioning and stabilizing structure (Figures 1 and 2, 2, "System 2 includes a pressure generating device 4, a delivery conduit 6, and a patient interface device 8." Column 4, Lines 45-50) to provide a force to hold a seal -forming structure (10, "patient interface device 8 includes a cradle style sealing cushion 10 coupled to a tubing assembly 12." Column 5, Lines 1-5) in a therapeutically effective position on a patient's head (Figures) for positive pressure airway treatment ("Pressure generating device 4 is structured to generate a flow of breathing gas and may include, without limitation, ventilators, constant pressure support devices (such as a continuous positive airway pressure device, or CPAP device), variable pressure devices (e.g., BiPAP®, Bi-Flex®, or C-Flex™ devices manufactured and distributed by Philips Respironics of Murrysville, Pa.), and auto-titration pressure support devices." Column 4, Lines 50-60) of sleep disordered breathing ("It is also known to deliver positive airway pressure (PAP) therapy to treat certain medical disorders, the most notable of which is OSA. .... a constant positive pressure is provided to the airway of the patient in order to splint open the patient's airway, and variable airway pressure, wherein the pressure provided to the airway of the patient is varied with the patient's respiratory cycle. Such therapies are typically provided to the patient at night while the patient is sleeping." Column 1, Lines 30-40), the positioning and stabilizing structure comprising: at least one gas delivery tube (14A/14B, "As seen in FIG. 1, tubing assembly 12 includes a first arm 14A

structured to rest along a first side of the patient's head and a second arm 14B structured to rest along an opposite second side of the patient's head when patient interface device 8 is donned by the patient." Column 5, Lines 5-10) to deliver a flow of air (via 4 of Figure 1) to an entrance of the patient's airways (nose as shown in Figure 1) via a seal forming structure (10, "cradle style sealing cushion 10"), each gas delivery tube (14A/14B) being constructed and arranged to contact, in use, at least a region of the patient's head superior to and otobasion superior of the patient's head (Figure 1), each gas delivery tube (14A/14B) comprising a tube wall (best seen Figures 3 and 4 - in cross sectional view) defining a hollow interior (24, "Referring now to FIGS. 2-4, each of first and second arms 14A and 14B of tubing assembly 12 includes a housing 22 which defines a passage 24 (FIGS. 3 and 4) therethrough through which a flow of breathing gas is conducted." (Column 5, Lines 30-35) through which air is able to flow to the seal forming structure (10), each gas delivery tube comprising: a first elongate length of material (30A/30B) having opposing first and second edges along its length; a second elongate length of material (one of 22 left or right of 32A/32B) having opposing first and second edges along its length; and a third elongate length of material (other of 22 left or right of 32A/32B) having opposing first and second edges along its length, wherein the first elongate length of material (30A/30B) is joined to a second elongate length of material (one of 22 left or right of 32A/32B) along a first edge of the first elongate length of material (30A/30B) and the second edge of second elongate length of material (one of 22 left or right of 32A/32B) thereby forming a first seam (defined where 30A/30B meet one 22), and the second elongate length of material (one of 22 left or right of 32A/32B) is joined (through 32A/32B) to the third elongate length of material (other of 22 left or right of 32A/32B) along the first edge of the second elongate length

of material (one of 22 left or right of 32A/32B) and the second edge of the third elongate length of material (other of 22 left or right of 32A/32B) thereby forming a second seam (defined where 32A/32B meet other 22), to form a tube wall of the gas delivery tube.

Yet, Grashow **does not expressly disclose** the configuration of "an outer layer of the gas delivery tube comprises a textile material and/or a foam material providing an outer surface of the first elongate length of material, the second elongate length of material and the third elongate length of material."

Ovizinsky teaches a similar gas delivery tube construction to Rinaldi for providing positive pressure to the airways of the patient using a seal forming structure.

Regarding the features of the "outer layer", Ovizinsky teaches the construction of an outer layer (1231/1232 - "the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224) made of a textile material ("spacer fabrics and textile cover portions … laminated fabric") providing an outer surface of the each of the first elongate length of material, the second elongate length of material and the third elongate length of material; whereby the resultant effect of this construction is "a crush resistant textile conduit." (Para 0224).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of Grashow to include the use of an outer layer as taught by Ovizinsky to impart crush resistance the gas delivery tube - thus, preventing occlusion of the tube which would limit/prevent the operation of the gas delivery tube to deliver gas to the patient.

As to Claim 2, the modified Grashow, specifically Grashow discloses each gas delivery tube (14A/14B) is configured to extend in use from a position on top of the patient's head, along a side of the patient's face and to a position adjacent the entrances to the patient's airways (best seen Figure 1).

As to Claim 9, the modified Grashow, specifically Grashow discloses each gas delivery tube (14A/14B) includes a patient contacting portion (via 30A/30B - "As shown in FIG. 2, each of first and second arms 14A and 14B also includes a first magnetic element 30A, 30B (shown schematically in dashed line) disposed on or in the patient facing side of arms 14A, 14B" Column 5, Lines 35-40) configured to contact the patient's body and a non-patient contacting portion (bounded by the 32A/32B - "a second magnetic element 32A, 32B (shown schematically in dashed line) disposed on or in the side of arms 14A, 14B which faces away from the patient." Column 5, Lines 35-40) configured to be oriented away from the patient's body in use, wherein the patient contacting portion includes the first elongate length of material (30A/30B) and the non patient contacting portion includes the second elongate length of material and the third elongate length of material (one of 22 left or right of 32A/32B OR other of 22 left or right of 32A/32B, respectively).

As to Claim 10, the modified Rinaldi, specifically Ovizinsky teaches an additional configuration wherein the gas delivery tube additionally has an inner portion constructed of a foam material and/or spacer material. In particular, Ovizinsky teaches a tube (1250) having an outer layer (1255) and an inner layer (1225) as best seen in Figure 54, wherein "The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or

fabric 1255 (e.g., synthetic or specified fabric))" (Para 0225) and "The tube 1225 is constructed of spacer fabrics including one or more outer layers 1225(1) (e.g., first and second layers) that cooperate to form the wall of the tube and an inner layer 1225(2) (e.g., third layer) supported within the internal passage provided by the one or more outer layers." (Para 0226); whereby the resultant effect of the additional inner layer is "an anti-crush or occlusion resistant structure". Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of the modified Rinaldi to include the use of an inner layer as taught by Ovizinsky to impart crush resistance the gas delivery tube - thus, preventing occlusion of the tube which would limit/prevent the operation of the gas delivery tube to deliver gas to the patient.

As to Claim 11, please see the rejection of Claim 1, the configuration of the outer layer on the gas delivery tube effectively provides the claimed sleeve configured to enclose each of the first elongate length of material, the second elongate length of material and the third elongate length of material.

As to Claim 12, please see the rejection of Claim 1, as each of the first elongate length of material (30A/30B), extend a different length along the gas delivery tube than the second elongate length of material and the third elongate length of material as shown in Figure 2 this construction would effectively yield the claimed configuration of each of the first elongate length of material, the second elongate length of material and the third elongate length of material having their own corresponding outer layers.

As to Claim 13, please see the rejection of Claim 1, where first seam (defined where 30A/30B meet one 22) and second seam (defined where 32A/32B meet other 22), the third

seam is the orientation by which the portion opposing side of 30A/30B meets the 22 of the third elongate length of material. *Applicant is reminded, the use of the transitional phrase "comprising" (MPEP § 2111.03) is open language and gives way to "additional, unrecited elements".*

As to Claim 14, the modified Grashow, specifically Grashow discloses the first elongate length of material (30A/30B) having a concave shape (Figure 3) to position the seams away from the patient's face in use.

As to Claim 20, the modified Grashow, specifically Grashow discloses a crown connector (16, "A second end 14A.sub.2 of first arm 14A and a second end 14B.sub.2 of second arm 14B are each fluidly coupled to a coupling connector 16 structured to rest on top of the head of the patient when patient interface device 8 is donned by the patient." Column 5, Lines 15-20) that is configured to be positioned on the top of the patient's head in use to fluidly connect between the pair of gas delivery tubes (14A/14B).

As to Claim 21, please see the rejection of Claim 1, which discloses the features of the seal forming structure (10) as positioned on the patient's head. Regarding the operational pressure of the gases supplied to the patient from the gas delivery tube via the seal forming structure, the modified Rinaldi, specifically Ovizinsky teaches the known operation pressures were known. In particular, "In use, the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient interface via the air delivery conduit." (Para 0105) to provide treatment of breathing conditions (Para 0002). Therefore, it would have been obvious to one having ordinary skill in the art to modify the operational pressure of the

modified Grashow to include the claimed at least 6 cm of water, as taught by Ovizinsky to be a known operational pressure suitable for imparting breathing treatments to the patient in need.

10. Claims 3-8 are rejected under 35 U.S.C. 103 as being unpatentable over Grashow (11,273,279 - 371 national stage entry of PCT/IB2017/056005 having a PCT publication date of April 12, 2018) in view of Ovizinsky et al. (2014/0102456) as applied to Claim 1 and further in view of Devens Jr. (7,166,099).

As to Claim 3, the modified Grashow, specifically Grashow discloses the features of each of the first elongate length of material (30A/30B), the second elongate length of material (one of 22 left or right of 32A/32B) and the third elongate length of material (other of 22 left or right of 32A/32B); yet, **does not expressly disclose** the configuration of the each of the elongate lengths having a plurality of layers.

Devens teaches a similar medical device to Grashow - in particular "medical tubing…and catheters" for the conveyance of fluid (liquid or gas) to/from the patient. Regarding the limitation of "plurality of layers", Devens teaches the construction of medical devices having multiple layers in order to impart "**variable or differential stiffness** along a length, e.g., the axial length, of the device." (Column 1, Lines 35-40), whereby the "medical device including **at least four layers** including a first material and a second material having a **different stiffness** than a stiffness of the first material, wherein at least one of the layers **varies in thickness** axially along the device." (Column 1, Lines 45-50) whereby the resultant effect of these various layer and thickness modifications results in a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines

25-35). As known stiffness is directly correlated to rigidity, and is diametrically opposed to flexibility/elasticity/extendibility.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the medical tubing of the modified Grashow to include the use of multiple layers as taught by Devens to impart a customized properties to prevent kinking, bucking, and damage to the medical device in use which would preclude or prevent the desired operation of the medical device to convey fluid (liquid/gas) to/from the patient.

As to Claim 4, the modified Grashow, specifically Devens teaches the configuration whereby the first elongate length of material has a different rigidity than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claim 5, the modified Grashow, specifically Devens teaches the configuration whereby the first elongate length of material has a greater thickness than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35).

Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction via increased thickness that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claims 6 and 8, the modified Grashow, specifically Devens teaches the configuration whereby the first elongate length of material has an increased number of layers at than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction via increased layers that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claim 7, the modified Grashow, specifically Devens teaches the configuration whereby the first elongate length of material has less extendibility/flexibility than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate

length of material necessitates a more rigid construction that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

11. Claims 15-18 are rejected under 35 U.S.C. 103 as being unpatentable over Grashow (11,273,279 - 371 national stage entry of PCT/IB2017/056005 having a PCT publication date of April 12, 2018) in view of Ovizinsky et al. (2014/0102456) as applied to Claim 1 and further in view of Rinaldi (2006/0180151).

As to Claim 15, the modified Grashow, specifically Grashow discloses the first seam (defined where 30A/30B meet one 22) and the second seam (defined where 32A/32B meet other 22); yet, **does not expressly disclose** the ability of the first and second seams to impart rigidity.

Rinaldi teaches a similar gas delivery device to Grashow, for delivering pressurized gas to the patient. Regarding the orientation of the first seam and second seam to impart rigidity, Rinaldi discloses the use of the first and second seams (30b/30d and 30c,30a, respectively, "four adjustment members 30a, 30b, 30c, 30d" Para 0041, as best seen in Figure 3d whereby "the adjustment member 30a is dimensioned to fit over the first and second fluid supply tubes 20, 15, and/or the first, second, third, and fourth supply tube segments 35, 40, 45, 50." Para 0057) to control the adjustment and conformability of the gas delivery tube about the patient's head. Explicitly, Rinaldi discloses "While four adjustment members 30a, 30b, 30c, 30d are shown in this figure, one skilled in the art will appreciate that fewer adjustment members could be used to allow the wearer to reposition the fluid supply tubes to achieve a more comfortable

position." (Para 0041). In particular, Rinaldi discloses the use of "a thin non-ferromagnetic metal ribbon or wire 65, 65' is incorporated along a long axis of the polymeric joint (adjustment member) 30a. The addition of the thin non-ferromagnetic metal ribbon or wire 65, 65' provides the necessary positioning memory not generally available in the inexpensive thermoplastic polymers normally found in nasal cannula constructions. " (Para 0061). In light of this construction the joints/seams between components are configured not only with an additional thickness which imparts a degree of rigidity but additionally with a rigidizing member which imparts even greater rigidity, for the purpose of providing a readily conformable shape suitable to meet the needs of the patient.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the seam/joint of the modified Grashow to include the construction of the adjustment members with or without the additional rigidizing structure, as taught by Rinaldi for the purpose of providing conformability of the gas delivery tube to the needs of the patient wherein the additional layer of material provides for rigidity in controlling the adjustment and the additional structure of the wire provides further rigidity in the shape memory of the gas delivery tube.

As to Claim 16, the modified Grashow, specifically Rinaldi discloses rigidizing element (65) may or may not be utilized, whereby Figures 4D/4E incorporate the rigidizing element; whilst, Figure 3D does not utilize the rigidizing element. Thus, the configuration as claimed whereby the first seam has a greater rigidity than the second seam can be a function of whether or not the rigidizing element was included within the seam.

As to Claims 17 and 18, please see the rejection of Claim 15, wherein modified Grashow, specifically Rinaldi discloses rigidizing element (65) is made from a "thin non-ferromagnetic metal ribbon or wire" in order to impart "positioning memory" (Para 0061).

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12. Claims 1, 2, 9, and 10-21 are rejected under 35 U.S.C. 103 as being unpatentable over Rinaldi (2006/0180151) in view of Ovizinsky et al. (2014/0102456) - *whereby each elongate length of material provides an in-series configuration of the gas delivery tube.*

As to Claims 1 and 19, Rinaldi discloses a positioning and stabilizing structure (Figures 1-1b) to provide a force to hold a seal-forming structure (55, "a nasal insufflating member 55" Para 0036) in therapeutically effective position on a patient's head for positive pressure airway treatment of sleep disordered breathing, the positioning and stabilizing structure comprising: at least one gas delivery tube (15/20, "a first fluid supply tube 20" Para 0036 and "a second fluid supply tube 15" Para 0037) to deliver a flow of air to an entrance of the patient's airways (via nose) via a seal-forming structure (55), each gas delivery tube (15/20) being constructed and arranged to contact, in use, at least a region of the patient's head superior to an otobasion superior of the patient's head (Figures 1-1b), each gas delivery tube (15/20) comprising a tube wall (defined by the interior lumen of 15/20) defining a hollow interior through which air is able to flow to the seal-forming structure (55), each gas delivery tube (15/20) comprising: a first elongate length of material (45/50, "first, second, third, and fourth supply tube segments 35, 40, 45, 50" Para 0040) having opposing first and second edges along its length; a second

elongate length of material (35/40, "first, second, third, and fourth supply tube segments 35, 40, 45, 50" Para 0040) having opposing first and second edges along its length; and a third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b) having opposing first and second edges along its length; wherein the first elongate length of material (45/50) is joined to the second elongate length of material (35/40) along a first edge of the first elongate length of material (45/50) and the second edge of the second elongate length of material (35/40) thereby forming a first seam (30b/30d, "four adjustment members 30a, 30b, 30c, 30d" Para 0041, as best seen in Figure 3d whereby "the adjustment member 30a is dimensioned to fit over the first and second fluid supply tubes 20, 15, and/or the first, second, third, and fourth supply tube segments 35, 40, 45, 50." Para 0057); and the second elongate length of material (35/40) is joined to the third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b) along a first edge of the second elongate length of material (35/40) and the second edge of the third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b) thereby forming a second seam (30a/30c - "four adjustment members 30a, 30b, 30c, 30d" Para 0041, as best seen in Figure 3d whereby "the adjustment member 30a is dimensioned to fit over the first and second fluid supply tubes 20, 15, and/or the first, second, third, and fourth supply tube segments 35, 40, 45, 50." Para 0057).

Yet, Rinaldi **does not expressly disclose** the configuration of "an outer layer of the gas delivery tube comprises a textile material and/or a foam material providing an outer surface of the first elongate length of material, the second elongate length of material and the third elongate length of material."

Ovizinsky teaches a similar gas delivery tube construction to Rinaldi for providing positive pressure to the airways of the patient using a seal forming structure.

Regarding the features of the "outer layer", Ovizinsky teaches the construction of an outer layer (1231/1232 - "the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube." Para 0224) made of a textile material ("spacer fabrics and textile cover portions … laminated fabric") providing an outer surface of the each of the first elongate length of material, the second elongate length of material and the third elongate length of material; whereby the resultant effect of this construction is "a crush resistant textile conduit." (Para 0224).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of Rinaldi to include the use of an outer layer as taught by Ovizinsky to impart crush resistance the gas delivery tube - thus, preventing occlusion of the tube which would limit/prevent the operation of the gas delivery tube to deliver gas to the patient.

As to Claim 2, the modified Rinaldi, specifically Rinaldi discloses each gas delivery tube (15/20) is configured to extend in use from a position on top of the patient's head, along a side of the patient's face, and to a position adjacent to the entrance of the patient's airways (Figures 1-1b).

As to Claim 9, the modified Rinaldi, specifically Rinaldi discloses the use of the first and second seams (30b/30d and 30c,30a, respectively, "four adjustment members 30a, 30b, 30c, 30d" Para 0041, as best seen in Figure 3d whereby "the adjustment member 30a is

dimensioned to fit over the first and second fluid supply tubes 20, 15, and/or the first, second, third, and fourth supply tube segments 35, 40, 45, 50." Para 0057) to control the adjustment and conformability of the gas delivery tube about the patient's head. Explicitly, Rinaldi discloses "While four adjustment members 30a, 30b, 30c, 30d are shown in this figure, one skilled in the art will appreciate that fewer adjustment members could be used to allow the wearer to reposition the fluid supply tubes to achieve a more comfortable position." (Para 0041). In light of this aforementioned conformability feature, the configuration by which the first elongate length of material contacts the patient closer than the remaining lengths of material is an obvious to try configuration from a finite number of predictable results having a reasonable expectation of success. As the first elongate length of material is proximate to the seal -forming structure, whilst, the remaining features elongate length of material are remote from the seal-forming structure the orientation by which the first elongate length of material acts to contact the patient; whilst the remaining features elongate length of material do not contact the patient as claimed is a likely configuration as the orientation of the remaining elongate length of material is a function of the location of the gas delivery source.

As to Claim 10, the modified Rinaldi, specifically Ovizinsky teaches an additional configuration wherein the gas delivery tube additionally has an inner portion constructed of a foam material and/or spacer material. In particular, Ovizinsky teaches a tube (1250) having an outer layer (1255) and an inner layer (1225) as best seen in Figure 54, wherein "The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric))" (Para 0225) and "The tube 1225 is constructed

of spacer fabrics including one or more outer layers 1225(1) (e.g., first and second layers) that cooperate to form the wall of the tube and an inner layer 1225(2) (e.g., third layer) supported within the internal passage provided by the one or more outer layers." (Para 0226); whereby the resultant effect of the additional inner layer is "an anti-crush or occlusion resistant structure". Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube of the modified Rinaldi to include the use of an inner layer as taught by Ovizinsky to impart crush resistance the gas delivery tube - thus, preventing occlusion of the tube which would limit/prevent the operation of the gas delivery tube to deliver gas to the patient.

As to Claim 11, please see the rejection of Claim 1, the configuration of the outer layer on the gas delivery tube effectively provides the claimed sleeve configured to enclose each of the first elongate length of material, the second elongate length of material and the third elongate length of material.

As to Claim 12, please see the rejection of Claim 1, as each of the first elongate length of material, the second elongate length of material and the third elongate length of material are separated by a seam the construction would effectively yield the claimed configuration of each of the first elongate length of material, the second elongate length of material and the third elongate length of material having their own corresponding outer layers.

As to Claim 13, please see the rejection of Claim 1, whereby the construction of the third seam (70) is the configuration of the other of the gas delivery tube (15/20) as connected through seal-forming structure (55) connected to the third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b) thus joining the first elongate length of

material (45/50) to the third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b).

As to Claim 14, the modified Rinaldi, specifically Rinaldi discloses first elongate length of material (45/50) having a concave shape as a function of the first seam (30b/30d) away from the patient's face. Rinaldi discloses "While four adjustment members 30a, 30b, 30c, 30d are shown in this figure, one skilled in the art will appreciate that fewer adjustment members could be used to allow the wearer to reposition the fluid supply tubes to achieve a more comfortable position." (Para 0041). Thus, the shape of the elongate length of material is a function of the shape of the adjustment member as readily modified by the patient to conform to the needs of the patient.

As to Claims 15-18, the modified Rinaldi, specifically Rinaldi discloses rigidizing element (65, "a thin non-ferromagnetic metal ribbon or wire 65, 65' is incorporated along a long axis of the polymeric joint (adjustment member) 30a." Para 0061) embedded within the first seam (30b/30d) and the second seam (30a/30c) to provide rigidity to the gas delivery tube. In particular, Rinaldi discloses the rigidizing element (65) "provides the necessary positioning memory not generally available in the inexpensive thermoplastic polymers normally found in nasal cannula constructions." (Para 0061). Thus, providing a customized fitment and adjustment retained for the use of the gas delivery tube by the patient in order "to achieve a more comfortable position" (Para 0041). Regarding Claim 16, the modified Rinaldi, specifically Rinaldi discloses rigidizing element (65) may or may not be utilized, whereby Figures 4D/4E incorporate the rigidizing element; whilst, Figure 3D does not utilize the rigidizing element. Thus, the configuration as claimed whereby the first seam has a greater rigidity than the second

seam can be a function of whether or not the rigidizing element was included within the seam. Regarding Claim 18, the modified Rinaldi, specifically Rinaldi discloses rigidizing element (65) is made from a "thin non-ferromagnetic metal ribbon or wire".

As to Claim 20, the modified Rinaldi, specifically Rinaldi discloses the crown connector (70, "A larger diameter common fluid supply tube 10 delivers the fluid from a reservoir to the first and second fluid supply tubes 20, 15 by way of a Y-fitting 70 depicted in FIG. 1b." Para 0040) for fluidly connecting between the pair of gas delivery tubes (15/20). Regarding the location of the crown connector (70) as shown in Figure 1B, the orientation of the crown connection is respective of the location of the gas source; thus if the source of gas was located above the head of the patient and/or the patient was laying down on a bed, the location of the crown connection would meet the claimed limitations of the crown connection "on top of the patient's head".

As to Claim 21, please see the rejection of Claim 1, which discloses the features of the seal forming structure (55) as positioned on the patient's head. Regarding the operational pressure of the gases supplied to the patient from the gas delivery tube via the seal forming structure, the modified Rinaldi, specifically Ovizinsky teaches the known operation pressures were known. In particular, "In use, the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient interface via the air delivery conduit." (Para 0105) to provide treatment of breathing conditions (Para 0002). Therefore, it would have been obvious to one having ordinary skill in the art to modify the operational pressure of the modified Rinaldi to include the claimed at least 6 cm of water, as taught by Ovizinsky to be a known operational pressure suitable for imparting breathing treatments to the patient in need.

13. Claims 3-8 are rejected under 35 U.S.C. 103 as being unpatentable over Rinaldi (2006/0180151) in view of Ovizinsky et al. (2014/0102456), as applied to Claim 1 and further in view of Devens Jr. (7,166,099).

As to Claim 3, the modified Rinaldi, specifically Rinaldi discloses the features of each of the first elongate length of material (45/50), the second elongate length of material (35/40) and the third elongate length of material (defined by the region beyond 30a/30c but before 70, Figure 1b); yet, **does not expressly disclose** the configuration of the each of the elongate lengths having a plurality of layers.

Devens teaches a similar medical device to Rinaldi - in particular "medical tubing…and catheters" for the conveyance of fluid (liquid or gas) to/from the patient. Regarding the limitation of "plurality of layers", Devens teaches the construction of medical devices having multiple layers in order to impart "**variable or differential stiffness** along a length, e.g., the axial length, of the device." (Column 1, Lines 35-40), whereby the "medical device including **at least four layers** including a first material and a second material having a **different stiffness** than a stiffness of the first material, wherein at least one of the layers **varies in thickness** axially along the device." (Column 1, Lines 45-50) whereby the resultant effect of these various layer and thickness modifications results in a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). As known stiffness is directly correlated to rigidity, and is diametrically opposed to flexibility/elasticity/extendibility.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the medical tubing of the modified Rinaldi to include the use of multiple layers as taught by Devens to impart a customized properties to prevent kinking, bucking, and damage to the medical device in use which would preclude or prevent the desired operation of the medical device to convey fluid (liquid/gas) to/from the patient.

As to Claim 4, the modified Rinaldi, specifically Devens teaches the configuration whereby the first elongate length of material has a different rigidity than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claim 5, the modified Rinaldi, specifically Devens teaches the configuration whereby the first elongate length of material has a greater thickness than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate

length of material necessitates a more rigid construction via increased thickness that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claims 6 and 8, the modified Rinaldi, specifically Devens teaches the configuration whereby the first elongate length of material has an increased number of layers at than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction via increased layers that the remaining elongate lengths of material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

As to Claim 7, the modified Rinaldi, specifically Devens teaches the configuration whereby the first elongate length of material has less extendibility/flexibility than the second elongated length of material and/or the third second elongated length of material as a function of the desire to provide a medical device with customized properties to be "more resistant to damage" and "less susceptible to kinking or buckling" during use (Column 2, Lines 25-35). Thus, the claimed configuration is obvious to try from a finite number of predictable results in order to address the concerns by which the expected forces experienced by the first elongate length of material necessitates a more rigid construction that the remaining elongate lengths of

material which are reliant on the placement of the gas delivery device and more distal from the patient's airways.

Conclusion

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNETTE F DIXON whose telephone number is (571)272-3392. The examiner can normally be reached M-F 9-5 EST with flexible hours.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kendra D Carter can be reached on 571-272-9034. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO

Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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